Highlights

Field accuracy is not heat accuracy: wall-flux-constrained surrogates for non-Boussinesq convection

- Field error below 1% leaves $> 2\%$ Nu error in 66% of 200 held-out samples.
- A wall-flux penalty cuts integral Nu error from 4.0% to 0.6%.
- Boussinesq–compressible gap factorises as $D = \varepsilon P(X)$ over 627 pairs.
- Single-contrast data are reusable within one boundary-condition family.
- Scaling law is the reusable object; rescaled plain matches within a few per cent.